

QUESTION 1. (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Evaluate $\int_0^3 \frac{x \, dx}{\sqrt{16 + x^2}}$.

3

(b) Find $\int \frac{dx}{x^2 + 6x + 13}$.

2

(c) Find $\int x e^{-x} \, dx$.

2

(d) Find $\int \cos^3 \theta \, d\theta$.

3

(e) (i) Find constants A , B and C such that

3

$$\frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} \equiv \frac{A}{1 + 2x} + \frac{Bx + C}{1 + x^2}.$$

(ii) Hence find $\int \frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} \, dx$.

2

(a) Given that $z = 1 + i$ and $w = -3$, find, in the form $x + iy$:

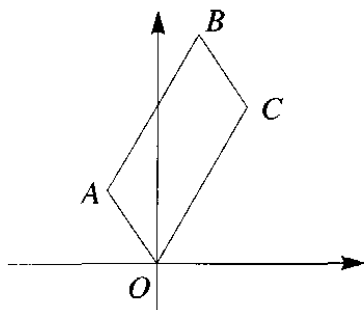
(i) wz^2 , 1

(ii) $\frac{z}{z + w}$. 2

(b) Using de Moivre's theorem, simplify $(-1 - i\sqrt{3})^{-10}$, expressing the answer in the form $x + iy$. 3

(c) Find the values of real numbers a and b such that $(a + ib)^2 = 5 - 12i$. 2

(d) 3



In the diagram above, $OABC$ is a parallelogram with $OA = \frac{1}{2}OC$.

The point A represents the complex number $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

If $\angle AOC = 60^\circ$, what complex number does C represent?

(e) z_1 and z_2 are complex numbers.

(i) Show that $|z_1| |z_2| = |z_1 z_2|$. 1

(ii) By taking $z_1 = 2 + 3i$ and $z_2 = 4 + 5i$, express 533 (the product of 13 and 41) as a sum of squares of two positive integers. 1

(iii) By taking other values for z_1 and z_2 , express 533 as a sum of squares of two other positive integers. 2

QUESTION 3. (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) On separate number planes, draw graphs of the following functions, showing essential features.

(i) $y = \frac{x+1}{x-1}$ 2

(ii) $y = \sqrt{\frac{x+1}{x-1}}$ 2

(iii) $y = \ln\left(\frac{x+1}{x-1}\right)$ 2

- (b) z is a variable complex number which is represented by the point P . Find the locus of P if $|z - 2i| = \text{Im}(z)$ 2

- (c) The fixed complex number α is such that $0 < \arg \alpha < \frac{\pi}{2}$. In an Argand diagram α is represented by the point A while $i\alpha$ is represented by the point B . z is a variable complex number which is represented by the point P .

(i) Draw a diagram showing A , B and the locus of P if $|z - \alpha| = |z - i\alpha|$. 1

(ii) Draw a diagram showing A , B and the locus of P if $\arg(z - \alpha) = \arg(i\alpha)$. 1

(iii) Find in terms of α the complex number represented by the point of intersection of the two loci in (i) and (ii). 1

- (d) Consider the function $y = \sin^{-1}(e^x)$.

(i) Find the domain and range of the function. 2

(ii) Sketch the graph of the function showing clearly the coordinates of any endpoints and the equations of any asymptotes. 2

QUESTION 4. (15 marks) Use a SEPARATE writing booklet.

Marks

Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$.

- (a) (i) Find the eccentricity of the ellipse. 1
 (ii) Find the coordinates of the foci and the equations of the directrices of the ellipse. 2
 (iii) Sketch the graph of the ellipse showing clearly all of the above features and the intercepts on the coordinate axes. 2
- (b) (i) Use differentiation to derive the equations of the tangent and the normal to the ellipse at the point $P(2,3)$. 3
 (ii) The tangent and normal to the ellipse at P cut the y axis at A and B respectively. Find the coordinates of A and B . 1
- (c) (i) Show that AB subtends a right angle at the focus S of the ellipse. 2
 (ii) Show that the points A, P, S and B are concyclic. 1
 (iii) Find the centre and radius of the circle which passes through the points A, P, S and B . 3

QUESTION 5. (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Let $P(x)$ be a degree 4 polynomial with a zero of multiplicity 3. Show that $P'(x)$ has a zero of multiplicity 2. 2
 (ii) Hence or otherwise find all zeros of $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$, given that it has a zero of multiplicity 3. 2
 (iii) Sketch $y = 8x^4 - 25x^3 + 27x^2 - 11x + 1$, clearly showing the intercepts on the coordinate axes. You do not need to give the coordinates of turning points or inflections. 1
- (b) (i) Show that the general solution of the equation $\cos 5\theta = -1$ is given by 2
 $\theta = (2n+1)\frac{\pi}{5}, \quad n = 0, \pm 1, \pm 2, \dots$
 Hence solve the equation $\cos 5\theta = -1$ for $0 \leq \theta \leq 2\pi$.
 (ii) Use De Moivre's Theorem to show that $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$. 3
 (iii) Find the exact trigonometric roots of the equation $16x^5 - 20x^3 + 5x + 1 = 0$. 2
 (iv) Hence find the exact values of $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}$ and $\cos \frac{\pi}{5} \cdot \cos \frac{3\pi}{5}$ and factorise 3
 $16x^5 - 20x^3 + 5x + 1$ into irreducible factors over the rational numbers.

- (a) A lifebelt mould is made by rotating the circle $x^2 + y^2 = 64$ through one complete revolution about the line $x = 28$, where all the measurements are in centimetres.

- (i) Use the method of slicing to show that the volume $V \text{ cm}^3$ of the lifebelt is given by **5**

$$V = 112 \pi \int_{-8}^8 \sqrt{64 - y^2} \, dy.$$

- (ii) Find the exact volume of the lifebelt. **2**

- (b) It is given that if a, b, c are any three positive real numbers, then $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$.

If $a > 0$, $b > 0$ and $c > 0$ are real numbers such that $a + b + c = 1$, use the given result to show that

(i) $\frac{1}{abc} \geq 27$ **1**

(ii) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$ **2**

(iii) $(1-a)(1-b)(1-c) \geq 8abc$ **2**

- (c) In a series of five games played by two equally matched teams, team A and team B, the team that wins three games first is the champion.

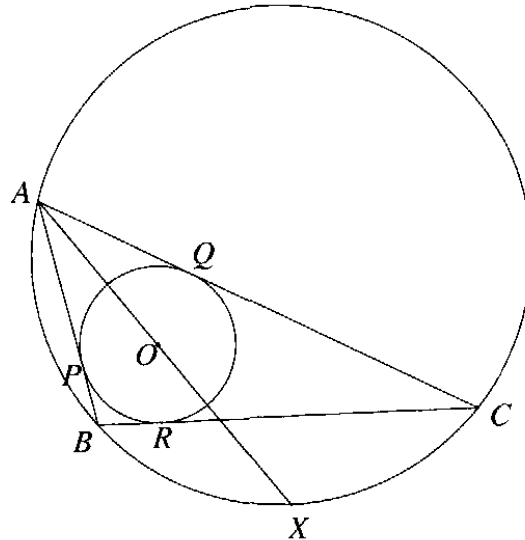
- (i) If team B wins the first two games, what is the probability that team A is the champion? **1**

- (ii) If team A has won the first game, what is the probability that team A is the champion? **2**

- (a) In the diagram below, ABC is a triangle.

The incircle tangent to all three sides has centre O , and touches the sides AB , AC and BC at P , Q and R respectively.

The circumcircle through A , B and C meets the line AO produced at X .



- (i) Show that $\angle CBX = \angle CAX$. 1
- (ii) Use congruence to prove that $\angle OBA = \angle OBC$. 2
- (iii) Prove that $\triangle XBO$ is an isosceles triangle. 3
- (iv) Prove that $BX = XC$. 1

- (b) (i) α . Differentiate $y = \log_e(1+x)$, and hence draw $y = x$ and $y = \log_e(1+x)$ on one set of axis. 1

- β . Using this graph, explain why 1

$$\log_e(1+x) < x, \text{ for all } x > 0.$$

- (ii) α . Differentiate $y = \frac{x}{1+x}$, and hence draw $y = \frac{x}{1+x}$ and $y = \log_e(1+x)$ on one set of axis. 1

- β . Using this graph, explain why 1

$$\frac{x}{1+x} < \log_e(1+x), \text{ for all } x > 0.$$

- (iii) Use the inequalities of parts (i) and (ii) to show that 4

$$\frac{\pi}{8} - \frac{1}{4}\log_e 2 < \int_0^1 \frac{\log_e(1+x)}{1+x^2} dx < \frac{1}{2}\log_e 2.$$

QUESTION 8. (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) At a dinner party there are twelve people, consisting of the six State Premiers and their partners. Each couple was representing one of the six States: New South Wales, Victoria, Western Australia, South Australia, Tasmania and Queensland.

- (i) The dinner took place at a circular table. Find how many seating arrangements are possible if:

α. there are no restrictions, 1

β. the males and females are in alternate positions. 1

- (ii) A committee of six is to be formed from the Premiers and their partners, where not more than one State can have two representatives. How many such committees are possible? 2

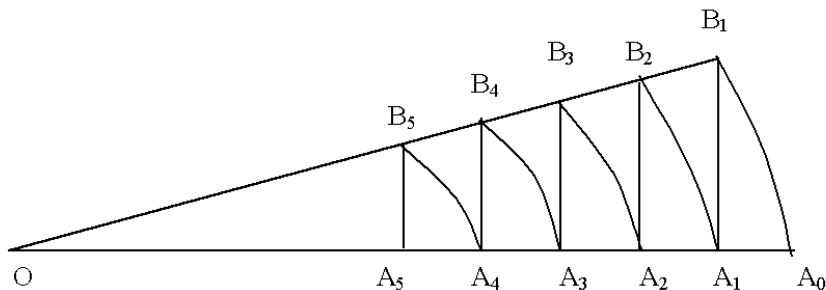
- (b) (i) Show that $\frac{t^n}{1+t^2} = t^{n-2} - \frac{t^{n-2}}{1+t^2}$. 1

- (ii) Let $I_n = \int \frac{t^n}{1+t^2} dt$. 1

Show that $I_n = \frac{t^{n-1}}{n-1} - I_{n-2}$, $n \geq 2$.

- (iii) Show that $\int_0^1 \frac{t^6}{1+t^2} dt = \frac{13}{15} - \frac{\pi}{4}$. 3

(c)



An ant walks along the circular arc from A_0 to B_1 , then down the straight line to A_1 , along the circular arc to B_2 , then down to A_2 , and so on, until it reaches O .

The length of OA_0 is 1, while angle A_0OB_1 is x radians, $0 < x \leq \frac{\pi}{2}$.

- (i) Show that the total distance the ant walks by the time it reaches O is given

by $y = \frac{x + \sin x}{1 - \cos x}$ 2

- (ii) Find the derivative of y with respect to x and explain why the derivative of y is always negative for all $0 < x \leq \frac{\pi}{2}$ 2

- (iii) Hence find the shortest possible distance the ant needs to walk from A_0 to O . 2

End of Paper

QUESTION 1

$$\begin{aligned}
 \text{(a)} \quad \int_0^3 \frac{x \, dx}{\sqrt{16+x^2}} &= \int_{16}^{25} \frac{\frac{1}{2} du}{\sqrt{u}} \\
 &= \frac{1}{2} \int_{16}^{25} u^{-\frac{1}{2}} du \\
 &= \frac{1}{2} \left[2u^{\frac{1}{2}} \right]_{16}^{25} \quad \checkmark \\
 &= \left[\sqrt{u} \right]_{16}^{25} \\
 &= \sqrt{25} - \sqrt{16} \\
 &= 5 - 4 \\
 &= 1 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } u &= 16 + x^2 \\
 du &= 2x \, dx \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \int \frac{dx}{x^2 + 6x + 13} &= \int \frac{dx}{(x+3)^2 + 4} \quad \checkmark \\
 &= \frac{1}{2} \tan^{-1} \frac{x+3}{2} + c \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad \int x e^{-x} dx &= \int x \frac{d}{dx} (-e^{-x}) dx \\
 &= -x e^{-x} - \int 1(-e^{-x}) dx \quad \checkmark \\
 &= -x e^{-x} + \int e^{-x} dx \\
 &= -x e^{-x} - e^{-x} + c \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \int \cos^3 \theta d\theta &= \int \cos^2 \theta \cos \theta d\theta \quad \checkmark \\
 &= \int (1 - \sin^2 \theta) \cos \theta d\theta \\
 &= \int (1 - u^2) du \quad \checkmark \\
 &= u - \frac{1}{3} u^3 + c \\
 &= \sin \theta - \frac{1}{3} \sin^3 \theta + c \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } u &= \sin \theta \\
 du &= \cos \theta d\theta
 \end{aligned}$$

(e) (i) Let $\frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} \equiv \frac{A}{1 + 2x} + \frac{Bx + C}{1 + x^2}$.

Then $x^2 - 4x - 1 \equiv A(1 + x^2) + (1 + 2x)(Bx + C)$ ✓

$$x^2 - 4x - 1 \equiv A + Ax^2 + Bx + C + 2Bx^2 + 2Cx$$

Equating coefficients of like terms,

$$1 = A + 2B \quad [\text{Eq. 1}]$$

$$-4 = B + 2C \quad [\text{Eq. 2}]$$

$$-1 = A + C \quad [\text{Eq. 3}]$$

Multiply Eq. 2 by -2:

$$8 = -2B - 4C \quad [\text{Eq. 2a}]$$

Eq. 1 + Eq. 2a:

$$9 = A - 4C \quad [\text{Eq. 4}]$$

Eq. 3 - Eq. 4:

$$-10 = 5C$$

$$C = -2$$

Substitute C into Eq. 3:

$$-1 = A - 2$$

$$A = 1$$

Substitute A into Eq. 1:

$$1 = 1 + 2B$$

$$B = 0 \quad \checkmark\checkmark$$

(ii)
$$\int \frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} dx = \int \left(\frac{1}{1 + 2x} + \frac{-2}{1 + x^2} \right) dx$$

$$= \frac{1}{2} \ln|1 + 2x| - 2 \tan^{-1} x + c \quad \checkmark\checkmark$$

QUESTION 2

(a) (i) $wz^2 = -3(1 + i)^2$

$$= -3(1 - 1 + 2i)$$

$$= -6i \quad \checkmark$$

(ii) $\frac{z}{z + w} = \frac{1 + i}{-2 + i} \times \frac{-2 - i}{-2 - i}$

$$= \frac{-(1 + i)(2 + i)}{5} \quad \checkmark$$

$$= \frac{-(1 + 3i)}{5}$$

$$= -\frac{1}{5} - \frac{3i}{5} \quad \checkmark$$

QUESTION 2

(b) $-1 - i\sqrt{3} = 2\text{cis}\left(-\frac{2\pi}{3}\right) \quad \checkmark$

Hence $\left(2\text{cis}\left(-\frac{2\pi}{3}\right)\right)^{-10} = 2^{-10}\text{cis}\left(\frac{20\pi}{3}\right) \quad \checkmark$

$$= 2^{-10}\text{cis}\left(\frac{2\pi}{3}\right)$$

$$= \frac{1}{1024}\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

$$= -\frac{1}{2048} + \frac{i\sqrt{3}}{2048} \quad \checkmark$$

(c)

$$(a + ib)^2 = 5 - 12i \Rightarrow (a^2 - b^2) + 2abi = 5 - 12i$$

$$\therefore a^2 - b^2 = 5 \quad \text{and} \quad ab = -6$$

$$a^4 - a^2b^2 = 5a^2 \Rightarrow a^4 - 5a^2 - 36 = 0$$

$$(a^2 + 4)(a^2 - 9) = 0 \quad \therefore a^2 > 0 \Rightarrow a^2 = 9$$

$$\therefore \begin{cases} a = 3 \\ b = -2 \end{cases} \quad \text{or} \quad \begin{cases} a = -3 \\ b = 2 \end{cases}$$

Criteria

- one mark for equating real and imaginary parts
- one mark for values of a and b

(d) To rotate $\vec{OA}$ by -60° , we need to multiply by $\text{cis}\left(-\frac{\pi}{3}\right) \quad \checkmark$.

Thus $\vec{OC} = 2 \times \vec{OA} \times \text{cis}\left(-\frac{\pi}{3}\right)$

$$= 2 \times \text{cis}\left(\frac{2\pi}{3}\right) \times \text{cis}\left(-\frac{\pi}{3}\right)$$

$$= 2\text{cis}\left(\frac{\pi}{3}\right) \quad \checkmark$$

$$= 1 + i\sqrt{3} \quad \checkmark$$

(e)

Criteria		Marks
(i)	• one mark for answer	1
(ii)	• one mark for answer	1
(iii)	• one mark for choice of z_1, z_2	2
(i)	• one mark for answer	

$$z_1 = a + ib, \quad z_2 = c + id \Rightarrow z_1 z_2 = (ac - bd) + i(ad + bc)$$

$$|z_1 z_2|^2 = (ac - bd)^2 + (ad + bc)^2 = a^2 c^2 - 2acbd + b^2 d^2 + a^2 d^2 + 2adbc + b^2 c^2$$

$$\therefore |z_1 z_2|^2 = (a^2 + b^2)(c^2 + d^2) = |z_1|^2 \cdot |z_2|^2 \quad \therefore |z_1| \cdot |z_2| = |z_1 z_2|$$

(ii)

$$z_1 = 2 + 3i \Rightarrow |z_1|^2 = 4 + 9 = 13$$

$$z_2 = 4 + 5i \Rightarrow |z_2|^2 = 16 + 25 = 41$$

$$z_1 \cdot z_2 = -7 + 22i \Rightarrow |z_1 \cdot z_2|^2 = 7^2 + 22^2$$

$$\therefore 533 = 13 \times 41 = 7^2 + 22^2$$

(iii)

For example :

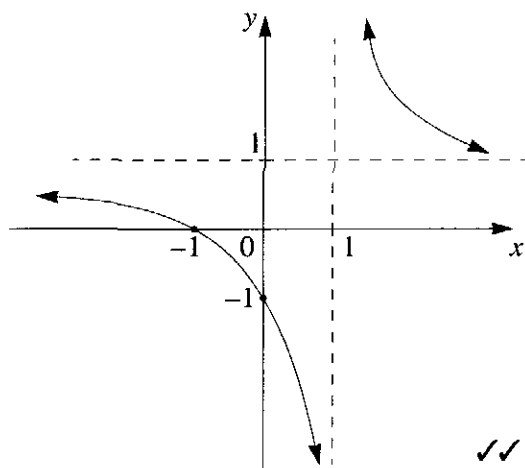
$$z_1 = 3 + 2i, \quad z_2 = 5 - 4i, \quad z_1 \cdot z_2 = 23 - 2i$$

$$|z_1|^2 = 13, \quad |z_2|^2 = 41, \quad |z_1 \cdot z_2|^2 = 23^2 + 2^2$$

$$\therefore 533 = 13 \times 41 = 23^2 + 2^2$$

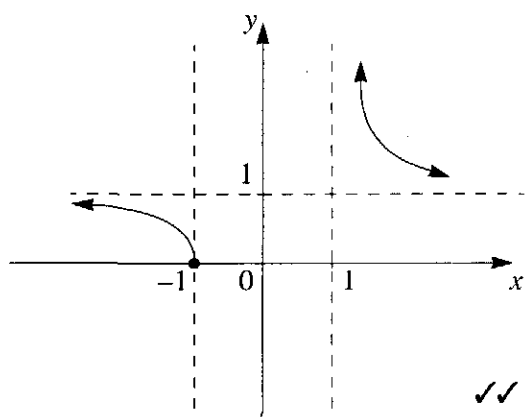
QUESTION 3

(a) (i)

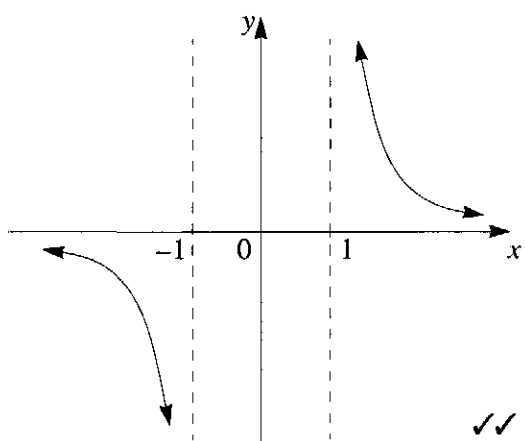


- vertical asymptote at $x = 1$
- horizontal asymptote at $y = 1$
- x intercept at $x = -1$
- y intercept at $y = -1$

(ii)



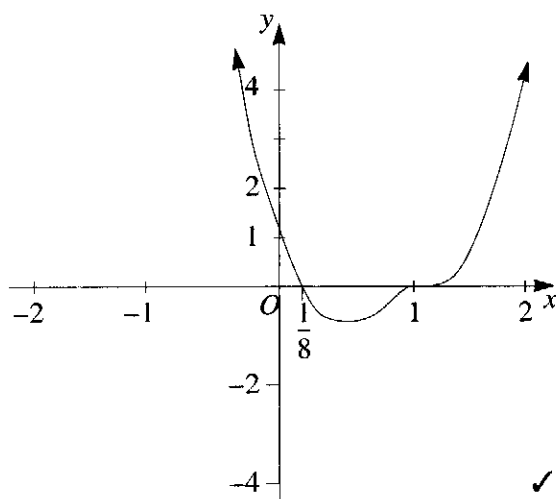
(iii)



- (b) The locus is the parabola with focus at $(0, 2)$ and the x axis as directrix.
or give the equation of the parabola as $x^2 = 4(y - 1)$

QUESTION 5

(a) (iii)



5(b)

Marking Guidelines

Criteria	Marks
(i) • one mark for general solution • one mark for particular solution	2
(ii) • one mark for expression for $\operatorname{Re}(\cos \theta + i \sin \theta)^5$ in terms of $\cos \theta$, $\sin \theta$ • one mark for expression for $\operatorname{Re}(\cos \theta + i \sin \theta)^5$ in terms of $\cos \theta$ • one mark for final answer	3
(iii) • one mark for noting that $x = \cos \theta$ where $\cos 5\theta = -1$ • one mark for solution	2
(iv) • one mark for value of $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}$ • one mark for value of $\cos \frac{\pi}{5} \cdot \cos \frac{3\pi}{5}$ • one mark for factorisation	3

Answer

(i)

$$\cos 5\theta = -1 \Rightarrow 5\theta = (2n+1)\pi$$

$$\theta = (2n+1)\frac{\pi}{5}, n = 0, \pm 1, \pm 2, \dots$$

$$0 \leq \theta \leq 2\pi \Rightarrow \theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

(iii) $16x^5 - 20x^3 + 5x + 1 = 0$

has solutions $x = \cos \theta$ where $\cos 5\theta = -1$.

$$x = \cos \frac{\pi}{5}, \cos \frac{3\pi}{5}, \cos \pi, \cos \frac{7\pi}{5}, \cos \frac{9\pi}{5}$$

$$x = \cos \frac{\pi}{5}, \cos \frac{\pi}{5}, \cos \frac{3\pi}{5}, \cos \frac{3\pi}{5}, -1$$

(iv)

$$\sum \alpha = 0 \Rightarrow 2\left(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}\right) - 1 = 0$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

Product of roots is $-\frac{1}{16}$

$$\therefore -\left(\cos \frac{\pi}{5} \cdot \cos \frac{3\pi}{5}\right)^2 = -\frac{1}{16}$$

$$\therefore \cos \frac{\pi}{5} \cdot \cos \frac{3\pi}{5} = -\frac{1}{4}$$

(since $\cos \frac{\pi}{5} > 0$, $\cos \frac{3\pi}{5} < 0$)

Then $\cos \frac{\pi}{5}$, $\cos \frac{3\pi}{5}$ are roots of

the equation $4x^2 - 2x - 1 = 0$. Hence

$$16x^5 - 20x^3 + 5x + 1 = (x+1)(4x^2 - 2x - 1)^2$$

(ii) Using the binomial expansion,

$$\operatorname{Re}\{(\cos \theta + i \sin \theta)^5\}$$

$$= \cos^5 \theta + 10 \cos^3 \theta (i \sin \theta)^2 + 5 \cos \theta (i \sin \theta)^4$$

$$= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2$$

$$= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

Using De Moivre's Theorem,

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

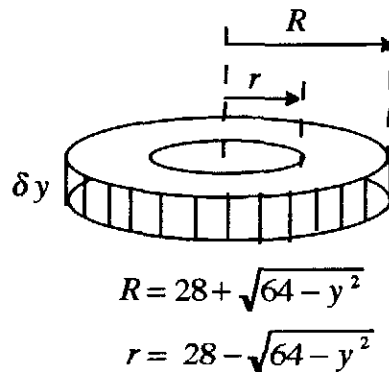
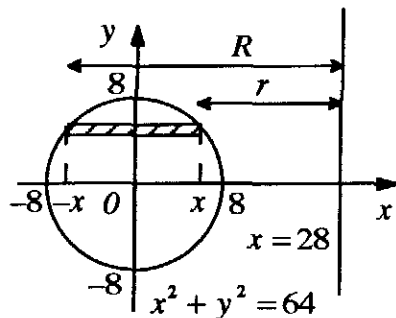
Hence $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$

Question 6

Marking Guidelines

(a)	Criteria	Marks
(i)	• one mark for identifying slice as annular prism, thickness δy • one mark for inner radius r in terms of y • one mark for outer radius R in terms of y • one mark for simplified value of δV in terms of y • one mark for expression for V	5
(ii)	• one mark for using area of semi circle, or appropriate integration process • one mark for final answer	2

(i)



Volume of slice is

$$\begin{aligned}
 \delta V &= \pi(R^2 - r^2)\delta y \\
 &= \pi(R+r)(R-r)\delta y \\
 &= \pi \cdot 56 \cdot 2\sqrt{64-y^2} \cdot \delta y \\
 V &= \lim_{\delta y \rightarrow 0} \sum_{y=-8}^8 112\pi\sqrt{64-y^2} \cdot \delta y \\
 &= 112\pi \int_{-8}^8 \sqrt{64-y^2} dy
 \end{aligned}$$

$$(ii) \int_{-8}^8 \sqrt{64-y^2} dy = \frac{1}{2}\pi \cdot 8^2 = 32\pi \quad (\text{Area of semicircle radius } 8) \Rightarrow V = 3584\pi^2$$

Exact volume of lifebelt is $3584\pi^2 \text{ cm}^3$

(b)

$$\begin{aligned}
 (i) \quad \text{RHS} &= t^{n-2} - \frac{t^{n-2}}{1+t^2} \\
 &= \frac{(1+t^2)t^{n-2} - t^{n-2}}{1+t^2} \\
 &= \frac{t^{n-2} + t^n - t^{n-2}}{1+t^2} \\
 &= \frac{t^n}{1+t^2} \\
 &= \text{LHS} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad I_n &= \int \frac{t^n}{1+t^2} dt \\
 &= \int \left(t^{n-2} - \frac{t^{n-2}}{1+t^2} \right) dt \\
 &= \frac{t^{n-1}}{n-1} - \int \frac{t^{n-2}}{1+t^2} dt \\
 &= \frac{t^{n-1}}{n-1} - I_{n-2} \quad \checkmark
 \end{aligned}$$

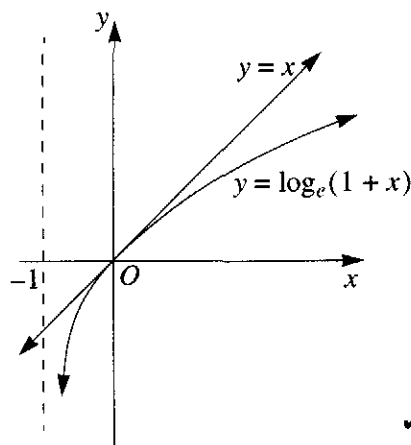
$$\begin{aligned}
 (iii) \quad \text{Let } J_n &= \int_0^1 \frac{t^n}{1+t^2} dt \\
 \text{Then } J_n &= \left[\frac{t^{n-1}}{n-1} \right]_0^1 - J_{n-2} \\
 &= \frac{1}{n-1} - J_{n-2} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence } J_6 &= \frac{1}{5} - J_4 \\
 &= \frac{1}{5} - \frac{1}{3} + J_2 \\
 &= \frac{1}{5} - \frac{1}{3} + 1 - J_0 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{But } J_0 &= \int_0^1 \frac{1}{1+t^2} dt \\
 &= \left[\tan^{-1} t \right]_0^1 = \frac{\pi}{4} \\
 \text{Hence } J_6 &= \frac{1}{5} - \frac{1}{3} + 1 - \frac{\pi}{4} \\
 &= \frac{13}{15} - \frac{\pi}{4} \quad \checkmark
 \end{aligned}$$

7 (b) (i) $\alpha.$ $y = \log_e(1+x)$

$$\frac{dy}{dx} = \frac{1}{1+x}$$

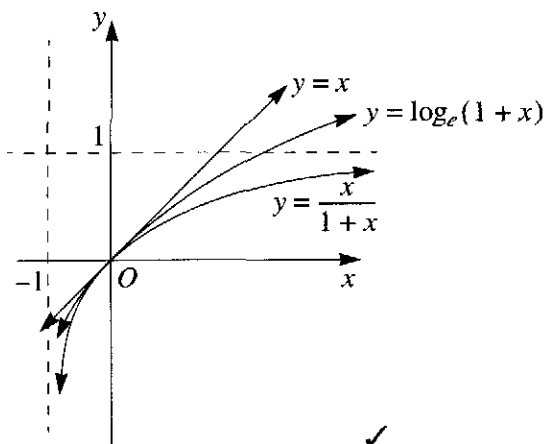


$\beta.$ When $x = 0$, $\frac{dy}{dx} = 1$, so $y = x$ is a tangent at $(0, 0)$.

Since $y = \log_e(1+x)$ is concave down, it follows that its graph is below the line $y = x$ for $x > 0$. ✓

(ii) $\alpha.$ $y = \frac{x}{1+x}$

Using the quotient rule, $\frac{dy}{dx} = \frac{1}{(1+x)^2}$.



$\beta.$ When $x = 0$, $\frac{dy}{dx} = 1$, so $y = x$ is a tangent to both curves at $(0, 0)$.

But for $x > 0$, the gradient function of $y = \frac{x}{1+x}$ is less than the gradient function of $y = \log_e(1+x)$, because $\frac{1}{(1+x)^2} < \frac{1}{1+x}$ for $x > 0$.

Hence the graph of $y = \frac{x}{1+x}$ is always below the graph of $y = \log_e(1+x)$ for $x > 0$. ✓

(iii) From (i) and (ii), $\frac{x}{1+x} < \log_e(1+x) < x$ for all $x > 0$.

Hence $\frac{x}{(1+x)(1+x^2)} < \frac{\log_e(1+x)}{1+x^2} < \frac{x}{1+x^2}$ for all $x > 0$

and so $\int_0^1 \frac{x}{(1+x)(1+x^2)} dx < \int_0^1 \frac{\log_e(1+x)}{1+x^2} dx < \int_0^1 \frac{x}{1+x^2} dx$ for all $x > 0$. ✓

$$\begin{aligned} \text{Now } \int_0^1 \frac{x}{1+x^2} dx &= \left[\frac{1}{2} \log_e(x^2 + 1) \right]_0^1 \\ &= \frac{1}{2} \log_e 2 \quad \checkmark \end{aligned}$$

7 (b) (iii)

(cont'd) Also, $\int_0^1 \frac{x}{(1+x)(1+x^2)} dx = \int_0^1 \left(-\frac{1}{2(x+1)} + \frac{1+x}{2(x^2+1)} \right) dx$ (partial fractions)

$$= \left[-\frac{1}{2} \log_e(1+x) \right]_0^1 + \left[\frac{1}{4} \log_e(x^2+1) \right]_0^1 + \left[\frac{1}{2} \tan^{-1} x \right]_0^1 \quad \checkmark$$

$$= -\frac{1}{2} \log_e 2 + \frac{1}{4} \log_e 2 + \frac{1}{2} \tan^{-1} 1$$

$$= \frac{\pi}{8} - \frac{1}{4} \log_e 2$$

Hence $\frac{\pi}{8} - \frac{1}{4} \log_e 2 < \int_0^1 \frac{\log_e(1+x)}{1+x^2} dx < \frac{1}{2} \log_e 2$ for all $x > 0$. $\checkmark$

QUESTION 8

- (a) (i) α . Number of arrangements when there are no restrictions = $11!$ $\checkmark$
 $= 39916800$
- β . The males and females are in alternate positions.
 Sit a person down. There are $5!$ ways of seating the remaining members of the same sex.
 Then there are $6!$ ways of seating the opposite sex.
 So the total number of ways = $5! \times 6!$ ways. $\checkmark$

(ii) Two cases:

(1) If one state has two representatives, number of ways = $\binom{6}{4} \times 2^5 = 480$ $\checkmark$

(2) If no state has two representatives, number of ways = $2^6 = 64$ $\checkmark$

Hence total number of ways = $480 + 64 = 544$

(b)	Criteria	Marks
(i)	• one mark for answer	1
(ii)	• one mark for replacing a, b, c by $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ respectively • one mark for final answer	2
(iii)	• one mark for use of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$ • one mark for final answer	2

(i) $\sqrt[3]{abc} \leq \frac{a+b+c}{3} = \frac{1}{3}$
 $abc \leq \frac{1}{27}$
 $\frac{1}{abc} \geq 27$

(ii) $\frac{1}{3} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq \sqrt[3]{\left(\frac{1}{a} \right) \left(\frac{1}{b} \right) \left(\frac{1}{c} \right)}$
 $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 \sqrt[3]{\frac{1}{abc}}$
 $\geq 3 \sqrt[3]{27}$
 $\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$

(iii) $(1-a)(1-b)(1-c)$
 $= 1 - (a+b+c) + (bc+ca+ab) - abc$
 $= (bc+ca+ab) - abc$
 $= abc \left\{ \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) - 1 \right\}$
 $\geq abc (9 - 1)$
 $\therefore (1-a)(1-b)(1-c) \geq 8abc$